

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-013 - Lakwa-013

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000148
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2012-12-07 / 2013-05-13
Measured / TVD	3037 m / 3021 m
Trajectory	Vertical; inc 5.7 deg; az 343.4 deg
Formation Interval	1386-2253 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	2902 psi
Fracture Gradient	15.47 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	Moderate (53.7/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	759	Surface
9 5/8 in casing	1761	Intermediate
7 in liner/casing	2672	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2013-01-28 1150 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Correlate pressure anomalies with flow before escalating risk.
2013-02-22 1609 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2013-03-17 2038 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2013-03-27 2207 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2013-03-29 2244 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2013-04-22 2671 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2012-12-27 534 m	Mud Loss	Medium	Flow-out dropped below flow-in with pit loss. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.

Completion Summary

The synthetic well reached 3037 m MD in the Barail interval. The NWIS knowledge record links 16 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.